

Euclid, Book I, Proposition I.14

If Two Adjacent Angles Are Together Equal to Two Right Angles, the Outer Sides Form One Straight

Line

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Euclid, Book I, Proposition I.14

Theorem 1 (Euclid I.14). *Let AB be a straight line, and let the straight lines BC and BD meet it at the point B on opposite sides of AB .*

If the adjacent angles

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are together equal to two right angles, then BC and BD form one straight line.

Equivalently,

$$C - B - D.$$

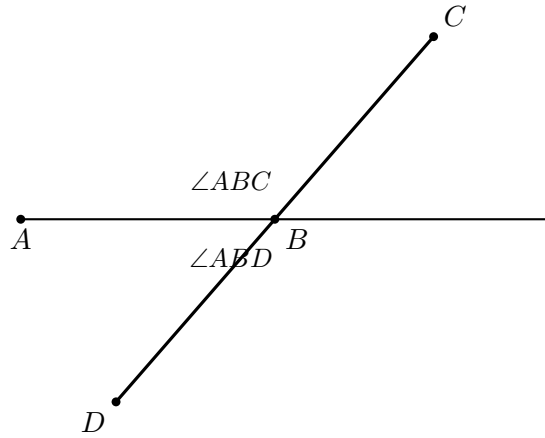


Figure 1: Euclid's configuration for Proposition I.14. The rays BC and BD lie on opposite sides of the straight line through A and B . If the adjacent angles $\angle ABC$ and $\angle ABD$ are together equal to two right angles, then C , B , and D lie on one straight line.

Step 1. Represent the sum of the two angles

The hypothesis that the two adjacent angles are together equal to two right angles is represented synthetically.

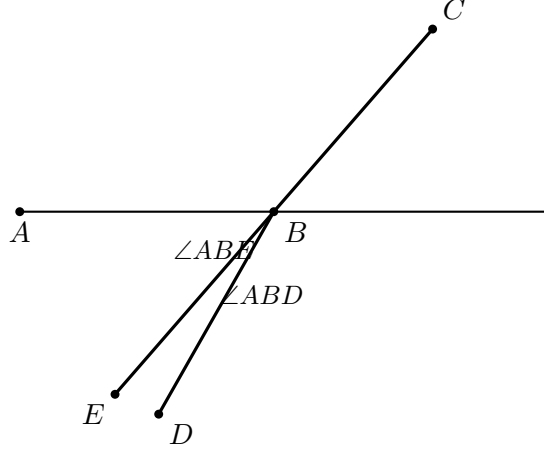
Choose a point E on the continuation of CB beyond B , so that

$$C - B - E.$$

Then $\angle ABE$ is the supplementary angle to $\angle ABC$. Consequently, the hypothesis of Proposition I.14 is expressed by

$$\angle ABD \cong \angle ABE.$$

Thus the formalization does not introduce a numerical measure of angles or an operation of angle addition. Instead, “equal to two right angles” is encoded by comparison with the angle determined by the opposite ray.



In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
[HilbertIncidence Geo]
(A B C D : Geo.Point) : Prop :=
exists E : Geo.Point,
Geo.Between C B E /\
Geo.AngleCongruent A B D A B E
```

Hence the witness E simultaneously records the opposite ray to BC and the angle congruence expressing the Euclidean hypothesis.

Step 2. Separate the collinear and noncollinear cases

The formal proof now distinguishes two possibilities for the three points C , D , and E .

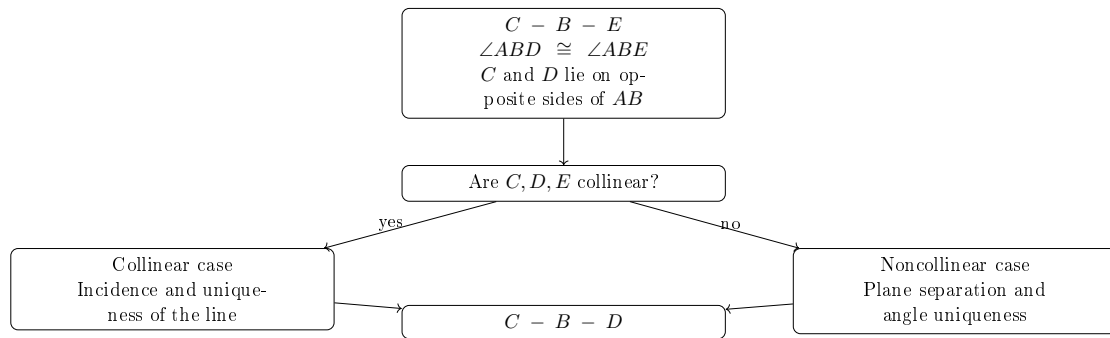
C, D, E are collinear

or

C, D, E are not collinear.

This division is not visible in Euclid’s original argument, but it is natural in the Hilbert reconstruction.

In the collinear case the conclusion follows from incidence and betweenness alone. In the noncollinear case the proof uses plane separation and the uniqueness part of Hilbert’s angle-construction axiom.



The two branches establish the same conclusion,

$$C - B - D,$$

but by substantially different parts of the Hilbert theory.

Step 3. The collinear case

Assume first that C , D , and E are collinear.

The point E was chosen so that

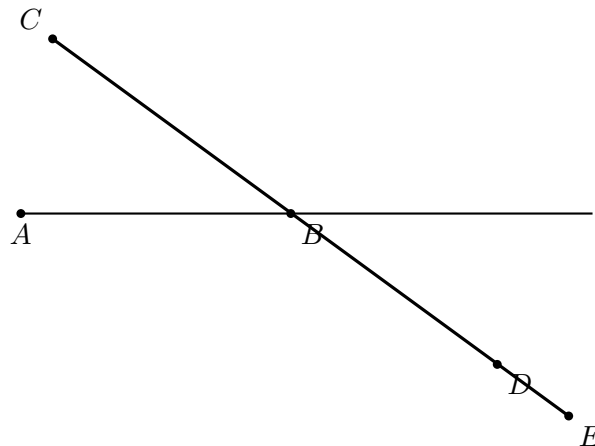
$$C - B - E.$$

Hence B lies on the same line as C and E .

On the other hand, since C and D lie on opposite sides of the base line AB , the segment CD meets the base line. Let X be a point of intersection, so that

$$C - X - D$$

and X lies on the line AB .



Both B and X therefore lie on the two lines:

$$AB$$

and

$$CDE.$$

If $X \neq B$, the uniqueness of the line through two distinct points would force the two lines to coincide. But then C would lie on the base line AB , contradicting the assumption that C and D lie on opposite sides of that line.

Hence

$$X = B.$$

Substituting this into

$$C - X - D$$

gives immediately

$$C - B - D.$$

Thus Proposition I.14 follows in the collinear case using only incidence, betweenness, and uniqueness of the line through two distinct points.

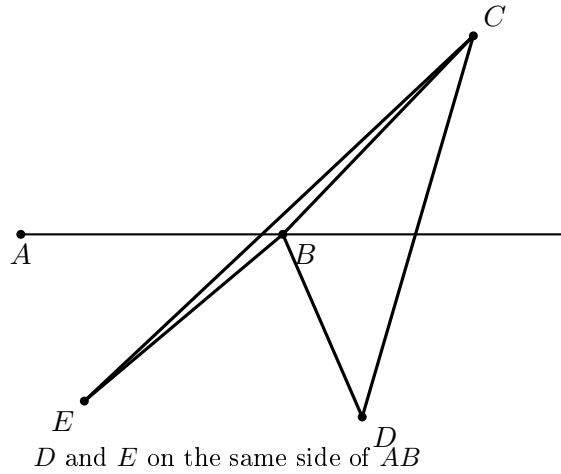
Step 4. The noncollinear case

Assume now that C , D , and E are not collinear.

The segment CD meets the base line AB , and the segment CE also meets the same line at B . Since CDE is a genuine triangle, the Hilbert plane-separation theorem implies that D and E lie on the same side of the base line.

Thus

$$\text{SameSide}(D, E; AB).$$



The angle $\angle ABE$ is nondegenerate. Hilbert's angle-construction axiom can therefore be applied at B on the side of AB containing D .

Construct a ray BZ such that

$$\angle ABZ \cong \angle ABE.$$

The uniqueness part of Hilbert III.4 now determines this ray. Since E lies on the same side of AB as D and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\text{SameRay}(B, Z, E).$$

The hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

Since D lies on the chosen side of AB , the same uniqueness clause also gives

$$\text{SameRay}(B, Z, D).$$

Therefore E and D determine the same ray from B :

$$\text{SameRay}(B, E, D).$$

In the Lean proof this last composition is supplied by the Book Zero ray theorem

`bookZero_36_ray3`

which composes two same-ray relations having a common reference ray.

Finally we already know

$$C - B - E.$$

Replacing E by another point D on the same ray from B preserves the betweenness relation. The Hilbert ray-transport theorem therefore yields

$$C - B - D.$$

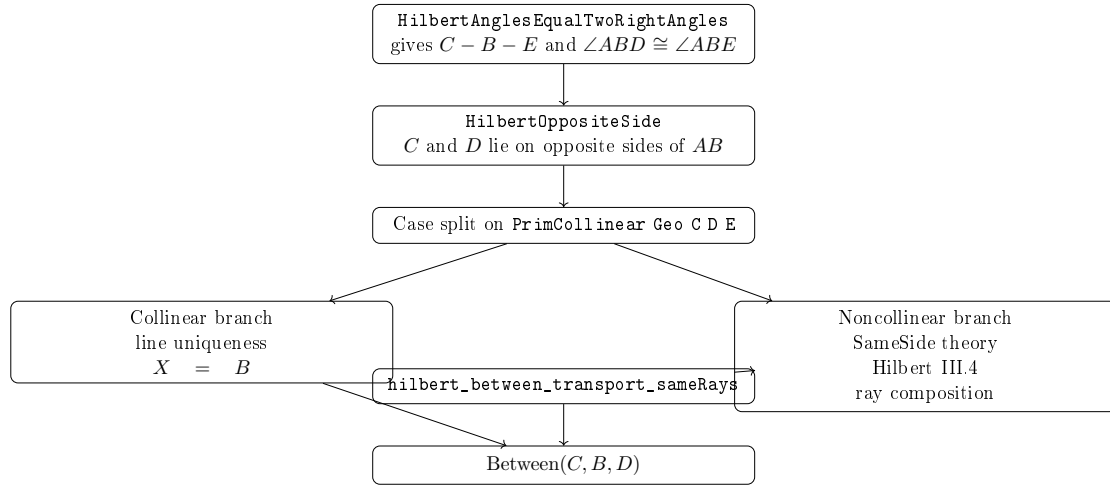
This completes the noncollinear case and hence the proof of Proposition I.14.

Lean Formalization

The final Lean theorem has the form

```
theorem euclid_proposition_14
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hOppCD : HilbertOppositeSide Geo C D base)
(hTwoRight :
HilbertAnglesEqualTwoRightAngles Geo A B C D) :
Geo.Between C B D := by
```

The formal dependency structure is



The proposition therefore does not require a new geometric axiom. Its only new interface notion is

HilbertAnglesEqualTwoRightAngles

which packages the Euclidean angle-sum hypothesis in purely synthetic Hilbert language.

The proof then combines already available incidence, order, separation, angle-construction, and ray machinery. In particular, the conclusion that BC and BD form one straight line is represented directly by

Between(C, B, D).

Reusable Hilbert and Book Zero ingredients

Several previously established results are used in the formal proof of Proposition I.14. Their role is worth recording explicitly.

hilbert_third_side_endpoints_sameSide. In the noncollinear branch the points C, D, E form a genuine triangle. The base line meets the interiors of the two sides CD and CE . The theorem

hilbert_third_side_endpoints_sameSide

then shows that the remaining endpoints D and E lie on the same side of the base line. Schematically,

$$\begin{array}{l} C - X - D, \\ C - B - E, \\ X, B \in AB \end{array} \quad \implies \quad \text{SameSide}(D, E; AB).$$

This is a derived plane-separation result based on Hilbert order theory.

Hilbert III.4: uniqueness of angle construction. Hilbert's angle-construction axiom does more than assert existence of a copy of a given angle. On a prescribed side of a prescribed ray, the constructed ray is unique.

In Proposition I.14 this uniqueness is applied twice. Both E and D lie on the same side of AB , and both realize an angle congruent to $\angle ABE$. Hence they must determine the same ray from B .

Thus

$$\text{SameRay}(B, Z, E)$$

and

$$\text{SameRay}(B, Z, D).$$

`bookZero_36_ray3.` The Book Zero theorem

`bookZero_36_ray3`

composes two same-ray relations with a common reference ray.
In the present proof,

$$\text{SameRay}(B, Z, E)$$

and

$$\text{SameRay}(B, Z, D)$$

yield

$$\text{SameRay}(B, E, D).$$

Thus the theorem acts as a small ray-calculus transitivity principle. It is one of the Book Zero results whose purpose becomes especially clear when used inside a later Euclidean proposition.

`hilbert_between_transport_sameRays.` The final step uses the theorem

`hilbert_between_transport_sameRays`

which transports strict betweenness when the two outer points are replaced by points on the same corresponding rays.

Here

$$C - B - E$$

is already known, the left ray BC is unchanged, and D lies on the same ray from B as E .
Therefore

$$C - B - E \quad \text{and} \quad \text{SameRay}(B, E, D)$$

imply

$$C - B - D.$$

This is the final synthetic step of Proposition I.14.

Conclusion

Proposition I.14 has been reconstructed entirely within the existing Hilbert geometry layer.

The Euclidean hypothesis that two adjacent angles are together equal to two right angles is represented synthetically by the predicate

`HilbertAnglesEqualTwoRightAngles`

through the existence of a point E satisfying

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

The proof then separates into two cases.

If C, D, E are collinear, incidence and line uniqueness identify the intersection of CD with the base line as B , giving immediately

$$C - B - D.$$

If C, D, E are noncollinear, plane separation places D and E on the same side of the base line. The uniqueness clause of Hilbert III.4 then shows that BD and BE are the same ray. Book Zero ray composition and transport of betweenness finally give

$$C - B - D.$$

Thus the formal conclusion is exactly

`Geo.Between C B D`

which is the strict-betweenness form of Euclid's assertion that BC and BD form one straight line.

No additional geometric axiom is required for Proposition I.14.